

Agentic Explainable Artificial Intelligence (Agentic XAI) Approach To Explore Better Explanation

AI for earth Observation Course - AI4EO - M2 GeoAI

Milestone 4

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Prepared by :

Manal Kabalan

Hasan Sarhan

Ali Ayach

Presented to: Dr Ali Ghandour

THE ORIGINAL PAPER



Original Paper: 66 Records,
High Validation R2 0.7487

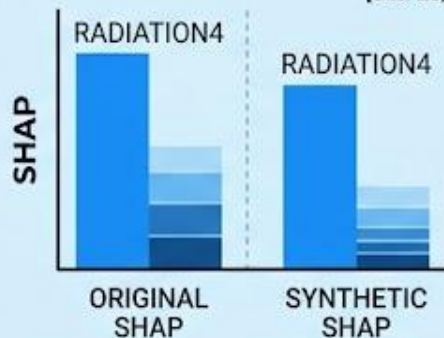
[cite: 19]



THE PREDICTIVE GAP

THE BRIGHT SIDE

XAI Workflow Reproduced:
Radiation4 as Dominant Feature
[cite: 20, 21]



Proof of Agentic XAI Workflow
Functionality [cite: 22, 24]

The Issue: Synthetic Model is Overfitted and
Inaccurate. Useful for Framework Testing [cite: 24],
But Not Validated For Deployment [cite: 23, 24]

THE SYNTHETIC BASELINE



Synthetic Baseline: 66 Records,
Low Validation R2 0.2215 [cite: 19]



**CLAIM A VALIDATED
RICE-YIELD PREDICTION
SYSTEM** [cite: 24]

Experiment 1: Stop-at-Round-4 Ablation [cite: 25]

Project: Agentic XAI workflow reproduction [cite: 3]



THE GOAL [cite: 27]

Test whether an iterative LLM refinement process benefits from running all the way to Round 10, or if stopping early at Round 4 yields a better result. [cite: 27]

Recommendation regeneration test must respond to evidence introduced earlier. [cite: 48]



THE FINDING [cite: 33, 63]

28/35
Progress math of recommence
Stopping at Round 4 was
the **superior**
approach, scoring 28/35 on
recommendation quality.



Rounds 5-10: Over-refinement [cite: 27]

WHY? [cite: 31, 32, 33, 36, 48, 60, 67, 70]

Rounds 5-10: Late Rounds

"global success?"
"large economic claims?"
"unvalidated assumptions?" [cite: 33, 36]

"are vague buzzwords?"
"vague words ..." [cite: 33, 36]

Round 4: Endpoint



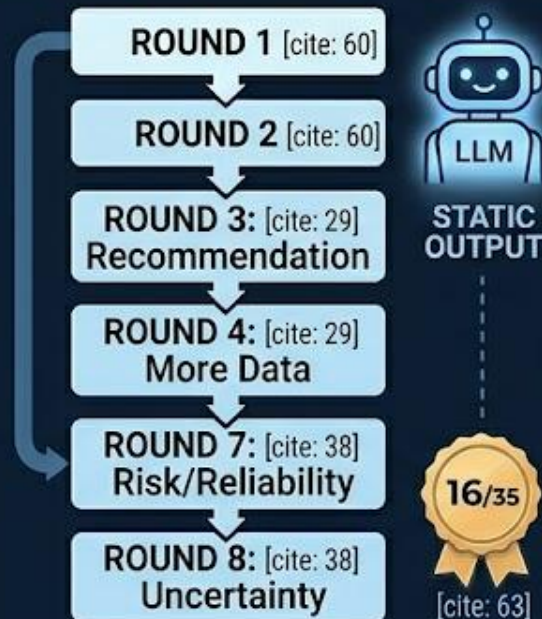
ENDPOINT: Cautious field pilot,
requests additional observations,
avoids promising fixed yield. [cite: 29]

Later rounds introduce assumptions not validated by available data [cite: 33]. **Early endpoint** clearer, more practical, and more closely connected to synthetic dataset [cite: 31, 67]. [cite: 31, 33, 67]

Experiment 2: Risk-First Order Sensitivity [cite: 37]

Project: Agentic XAI workflow reproduction [cite: 3]

ORIGINAL ORDER (Baseline) [cite: 60]



? WHY? [cite: 42, 60, 68]

1 'STATIC EVIDENCE' [cite: 48, 68]



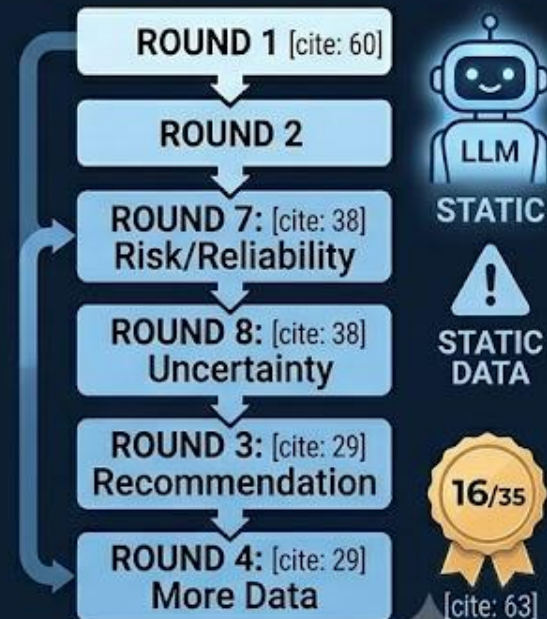
Analytic evidence (Risk/Reliability/Uncertainty) was static and did not **regenerate** after each round [cite: 48]. Changing order was insufficient [cite: 48].

2 'NO DYNAMIC REASONING' [cite: 48]



The language model reasoning did not update or respond to evidence introduced earlier [cite: 48]. Endpoint retained identical assumptions [cite: 44].

RISK-FIRST ORDER (Test)

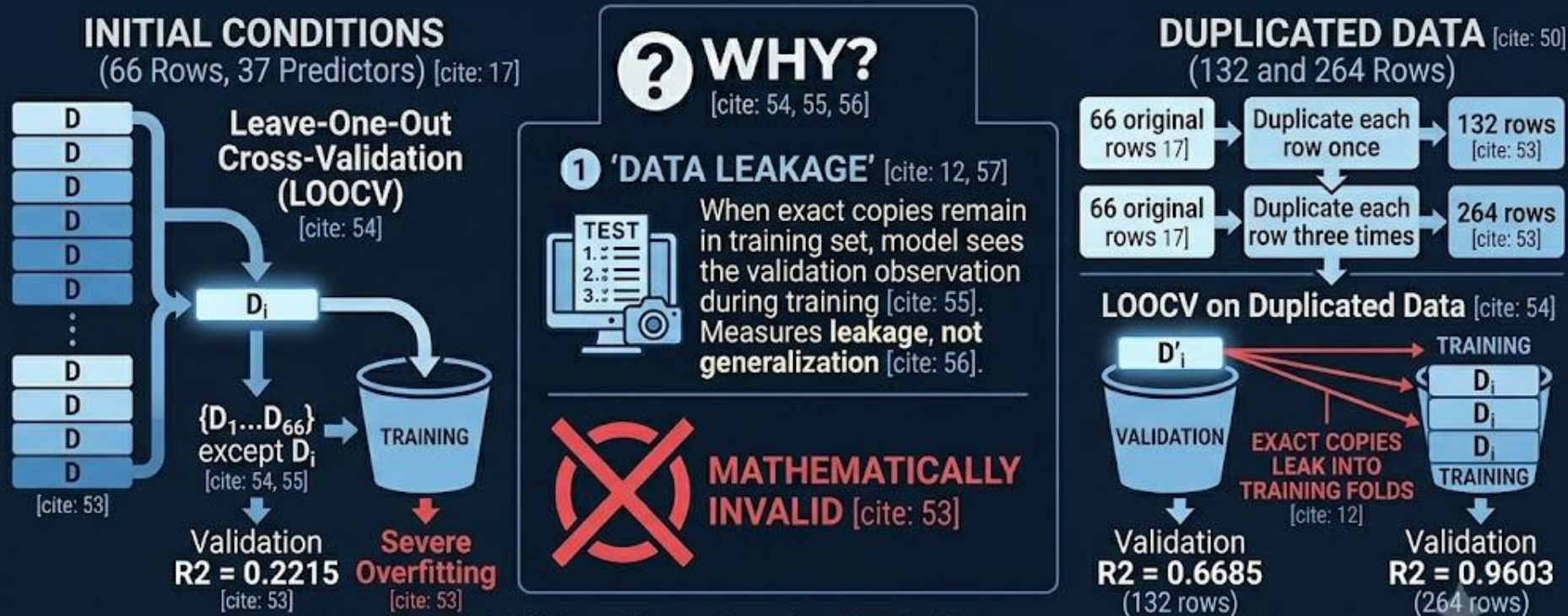


The Experiment Did Not Demonstrate Risk-First Ordering is Better [cite: 46].

Reordering static analytical evidence had **NO** meaningful change on the recommendation [cite: 60].

Experiment 3: Duplicate-Data Validation Stress Test [cite: 49]

Project: Agentic XAI workflow reproduction [cite: 3]



Experiment Did Not Show Genuine Model Improvement [cite: 52].

The Rise in R^2 Scores is Due to Contamination from Duplicates [cite: 53, 56].



SCIENTIFICALLY VALUABLE DEMONSTRATION: Confirms Why Duplicate Detection is Essential Before Evaluation [cite: 57, 58].

UNLOCKED METHODOLOGICAL INSIGHTS [cite: 70]

Project: Agentic XAI workflow reproduction [cite: 7]



GENAI & LLM-DRIVEN RECOMMENDATION QUALITY [cite: 67]

MORE ROUNDS DO NOT MEAN BETTER OUTPUTS [cite: 71]: **OVER-REFINEMENT** INTRODUCES **UNVALIDATED ASSUMPTIONS** [cite: 33].



Ultimately, **Recommendation Quality** relies on **strict evidence tracking** [cite: 70], and **Predictive Quality** relies on **leakage-free validation** [cite: 70].

Thank you. Ready to move on to **TASK 4**. [cite: user input]

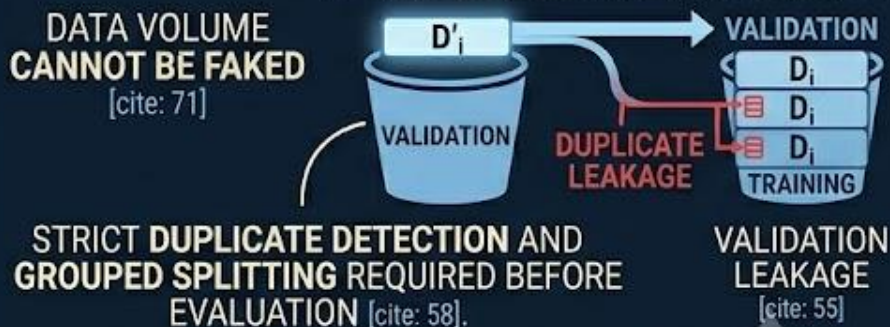


PREDICTIVE QUALITY & DATA LEAKAGE [cite: 70]

COPIED ROWS MERELY CREATE **VALIDATION LEAKAGE, NOT GENUINE MODEL GENERALIZATION** [cite: 12, 78].

DATA VOLUME CANNOT BE FAKED [cite: 71]

LEAKAGE-FREE VALIDATION DESIGN



Milestone 4: Generalizing Agentic XAI from Rice Yield to Wildfire Prediction

Task 3:
Rice-yield regression



Task 4:
Wildfire classification

Goal:
Transform a rice-specific workflow into a generalized Agentic XAI framework.



Generalized Workflow

Load Wildfire Dataset



Spatial Train-Test Split



Train Random Forest



SHAP Explainability



4 Refinement Rounds



Generate Recommendations

- **Dataset:**
 - 173,991 samples
- **Model:**
 - Random Forest (300 trees)
- **Explainability:**
 - SHAP

Agentic XAI Process

Iterative Refinement Rounds

Round 1



Generalization & Calibration

- Detect overfitting

Round 2



Subgroup Analysis

- Find terrain failures

Round 3



Feature Relationships

- Detect redundant variables

Round 4



Robustness Analysis

- Stress-test features

Main Findings & Conclusion ★

Key Findings

✗ Severe overfitting
99.9% → 58.4%

✗ Low fire detection
Recall = 38.9%

✗ Feature redundancy
Elevation variables overlap

✗ Aspect issue
Aspect hurts generalization

✓ Generalized from
rice to wildfire

✓ Identified model
weaknesses

✓ Generated
evidence-driven
analyses

✓ Suggested future
improvements

The workflow succeeded, even though the wildfire model failed.

AGENTIC XAI WORKFLOW FOR ENERGY CONSUMPTION PREDICTION



STEP 1

DATA PREPARATION AND FEATURE ENGINEERING



Dataset:
energydata_complete.csv



Target variable:
Appliances



Removed:

- date
- rv1
- rv2



Created time-based features:

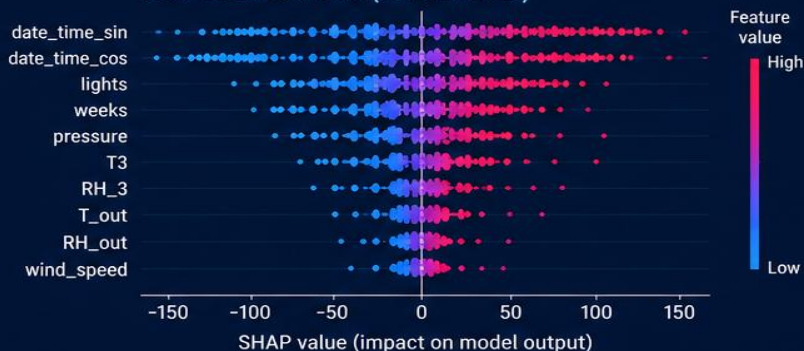
- date_time_sin
- date_time_cos
- weeks
- months
- weekend

STEP 2

BASELINE TRAINING AND EXPLAINABILITY

Component	Value
Model	Random Forest Regressor
Trees	300
Split	Random 80/20
R ² Score	0.6171

SHAP BEESWARM PLOT (TOP FEATURES)



We trained a Random Forest model using a random split. Initially, the performance looked good with an R^2 of 0.6171.



SHAP revealed that time-related features and lighting conditions were among the most influential variables.

Validation Stress Test (Most Important Slide)

Step 3: Validation Stress Testing

Validation Strategy	R ² Score
Random Split	0.6171
Grouped-Date Split	-0.0527
Future Temporal Split	-3.6599

- This is the turning point of the framework.
- When we tested the model under realistic conditions, performance collapsed.
- The random split created temporal leakage because observations recorded every 10 minutes were extremely similar.
- The model was memorizing recent events rather than forecasting future consumption.



Step 4: Subgroup Error Analysis

Highest error detected:

- Condition: lights = 30
- MAE = 50.43



- The framework automatically searched for situations where predictions fail.
- The largest errors occurred during **active household periods** when lights were on.
- This suggests that irregular human behavior is difficult for the model to predict.

Step 5: Feature Behavior Analysis

Features | Spearman Correlation

date_time_sin ↔ date_time_cos



0.4315

- The framework examined how features behave together.
- Time-based variables showed strong co-movement.
- However, correlation does not imply causation; this is only an indicator of related behavior.

Step 6: Robustness and Explanation Stability



Feature Permutation

- Permuting date_time_cos
- R² dropped by 0.4394



Stability Analysis

- Some features (e.g. T6) disappeared from the top rankings

- The framework then tested how dependent the model was on each feature.
- The model was highly dependent on **time-of-day variables**.
- Some features were unstable, meaning their importance changed depending on the data subset.

Human-in-the-Loop Agentic Output

Step 7: LLM Prompt and Evidence Generation



The system automatically generates:

Step 7 Generation	Output Types
Evidence files	Detailed data files for model inputs and outputs
Performance metrics	Comprehensive system performance data (e.g., Accuracy, F1-Score)
Error tables	Breakdown of specific errors and edge cases
Permutation results	Effects of input variations on the model
Structured prompts for LLMs	Context-aware prompts for language models



- The final stage does not directly call an LLM. 
- Instead, it prepares structured evidence that can later be given to an LLM to generate human-readable explanations for end users such as building managers. 

Key Takeaways

Key Findings

1

✓ Good performance with random split can be misleading.



2

✓ Temporal leakage severely affects reliability.



3

✓ Human behavior is difficult to predict.



4

✓ Agentic XAI continuously audits and explains model decisions.



Thank you!